

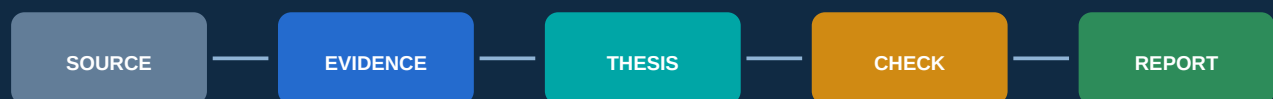
Evidence-Grounded Equity Research Agent

Final implementation, architecture, evaluation, reliability, and future-state design

A step-by-step account of how one ticker question becomes a balanced, cited bull case, bear case, risk analysis, financial view, and controlled release decision.

Prepared 7 September 2026

Case study: NVDA



Executive overview

This project addresses a familiar research problem: public-company information is abundant, but a useful investment research brief requires more than collecting documents. An analyst must define the reporting window, reconcile numbers, retrieve the right filing and transcript passages, construct competing explanations, test those explanations, cite every factual premise, and stop when evidence is too weak. The intended user is a retail investor, finance student, or junior analyst who wants a disciplined research starting point rather than a personalized trade recommendation.

The implemented system is a twelve-node LangGraph workflow with five explicit model-assisted roles: Planner, Researcher, Thesis Analyst, Critic, and Editor. Deterministic code surrounds those roles. It validates the request, resolves a common date window, prepares documents, calculates financial metrics, checks evidence coverage, prunes overlapping or weak theses, runs report-level guardrails, decides whether human review is required, and finalizes or withholds the report. The result is not a single unexamined answer from a language model; it is an auditable workflow with bounded return paths.

The operating principle is: required evidence → targeted research → balance check → Analyst → Critic → one refinement → deterministic report → quality gate → finalize or review.

The current source cache already avoids repeated downloading, cleaning, chunking, and embedding of complete SEC filings and earnings-call transcripts. It does not cache a final answer. Repeating the same question therefore still refreshes current data and reruns planning, retrieval, thesis generation, criticism, and release checks. A future exact-report cache can reduce an immutable historical repeat query to a freshness check plus retrieval, but it should be separate from semantic vector search and must be versioned by source, market snapshot, model, prompt, schema, and application code.

The user-facing workflow uses one balanced analysis path with a bounded Analyst-Critic refinement. Thesis presentation is organized by business theme, while strengthening conditions, monitored indicators, and invalidation conditions are explained once at the thesis level in complete sentences. Deterministic text normalization handles common HTML and PDF extraction artifacts. Topic controls also prevent two same-side theses from occupying separate slots when both depend on the same primary business driver.

Reading map

The report begins with the user problem and scope, then follows the system in the same order that a query moves through the graph. After the architecture and node-level walkthrough, it explains the data and RAG design, shows an NVDA execution, documents evaluation and observability, and closes with safety, design principles, limitations, and the future roadmap.

- 1. Problem, user, goal, and scope
- 2. Solution and overall architecture
- 3. LangGraph workflow and node walkthrough
- 4. Tools, RAG, memory, and repeated-query caching
- 5. NVDA case study
- 6. Evaluation methods and results
- 7. Observability, safety, and human oversight
- 8. Design principles, testing, limitations, and roadmap

Current capability	What it means
One balanced mode	One bounded Analyst-Critic refinement is used by default; no end-user mode choice remains.
Persistent source RAG	Complete filings and transcripts are reused from Chroma instead of downloaded and embedded again.
Cited cases	Bull, bear, and risk claims retain evidence objects and reader-facing source links.
Release control	A deterministic quality decision passes, repairs, or escalates the draft before finalization.
Verified local build	315 automated tests cover graph behavior, evidence handling, evaluation, reporting, and review paths.

1. Problem, intended user, goal, and scope

Equity research is a synthesis problem under uncertainty. Financial statements provide precise historical numbers but limited explanation. Earnings calls provide management context but are selective and forward-looking. News and analyst ratings add recency and external perspective but can be noisy. A direct language-model answer can blend these sources fluently without preserving the boundary between verified fact, calculation, and interpretation.

The system is designed for a user who understands that research supports a decision but does not make the decision. It helps the user answer questions such as how margins are changing, which operational driver could sustain growth, what evidence supports the downside case, and which observation would invalidate a thesis. It does not optimize a portfolio, predict a price, choose a trade, or tailor advice to a person's financial circumstances.

The project goal is to produce a professional, balanced, and traceable company research report from a ticker and a question. The scope covers public US-company information: SEC filings, earnings-call transcripts, market financial history, company profile data, current news, and analyst recommendation distributions. The analysis uses one resolved period window so that text and numbers refer to the same reporting context.

Success is not defined as sounding confident. Success means that the system either publishes a balanced report whose factual premises can be traced, repairs a correctable evidence gap, or pauses for a human when the release conditions are not satisfied.

2. The solution: an evidence-to-decision workflow

The implementation separates four kinds of work. Evidence acquisition gathers source documents and numerical data. Reasoning turns that evidence into alternative explanations. Assurance checks evidence support, completeness, and behavioral risk. Presentation turns the approved state into a readable report with charts, business-theme sections, full-sentence thesis evaluation guidance, and a separate linked source appendix.



Required evidence means the Planner first defines what must be known, not what answer should be written. Targeted research executes one plan task at a time with the appropriate tool. The balance check looks for missing positive, negative, financial, transcript, or risk evidence before thesis generation. The Analyst proposes grounded angles; the independent Critic applies a citation gate and a four-part scoring rubric. One bounded refinement gives the Analyst a chance to deepen-not endlessly rewrite-the surviving sides.

The Editor is deterministic in structure: it assembles already selected evidence, calculations, thesis fields, confidence, open questions, and source objects into a validated schema. The quality gate then combines report completeness, groundedness, confidence, source checks, and monitoring signals into one routing decision. Pass goes to finalization, repair returns only the missing or unsupported task to research, and review pauses the graph.

Overall system architecture

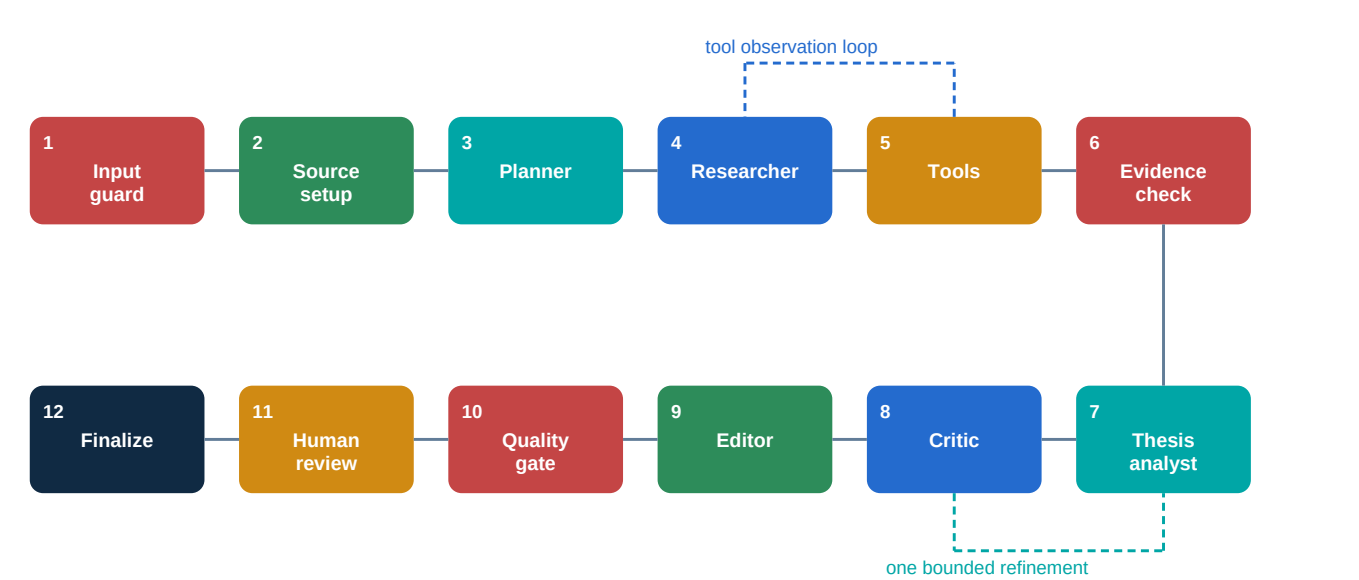


One shared blackboard connects every layer; every branch and decision is traceable.

The architecture is deliberately layered, but the layers do not exchange informal chat messages. Every node reads and writes a typed shared state-the graph blackboard. Reducers append evidence, trace steps, and messages across iterations; ordinary fields use last-write-wins semantics. This design makes a branch reproducible from the state that existed when it was evaluated and makes ownership of each control decision visible in a trace.

LangGraph supplies state-machine orchestration, conditional edges, the Researcher-ToolNode action loop, checkpointing, and interruption/resume for human review. LangChain supplies model bindings, message types, structured outputs, tool definitions, embeddings, text splitters, and the Chroma integration. LangSmith is optional operational telemetry: when enabled with an API key, it records graph runs, tags, timing, and safe metadata such as ticker, as-of value, request type, and thread identifier; raw credentials are excluded from trace metadata.

3. The twelve-node LangGraph



Conditional routes also send missing evidence or unsupported report claims back to research; a reviewer-requested revision returns to planning. Every return path is bounded by a counter.

The graph has exactly twelve top-level nodes. The five individual research actions are tools inside one ToolNode; they are not extra graph nodes. This keeps traces readable and prevents an implementation detail-adding a new tool-from changing the public workflow topology. Three bounded back-edges provide controlled correction: tools return an observation to the Researcher, the Critic can request one thesis refinement, and the quality gate or reviewer can return a targeted issue to research or planning.

Route	Condition	Outcome
Researcher → Tools → Researcher	The next plan task needs a data or retrieval action.	The ToolNode executes the action and returns a structured observation.
Evidence check → Researcher	A required evidence role is missing and the retry budget remains.	Only the gap is reopened; collected evidence stays on the blackboard.
Critic → Thesis Analyst	Balanced depth has not reached its configured cap.	One refinement deepens surviving cases; then the Critic scores again.
Quality gate → Researcher	The draft contains repairable unsupported or incomplete content.	A bounded report-repair task is created.
Quality gate → Human review	A hard trigger or monitor escalation requires judgment.	The graph checkpoints and waits for allow, revise, or block.

Nodes 1-3: define a safe and coherent research job

- 1. Input guard.** The first node validates the ticker and question, detects advice-seeking or unsafe input, and can reroute a request from personalized advice to neutral research. Invalid input stops immediately. This is a deterministic boundary: no expensive source or model work begins until the request is in scope.
- 2. Source setup.** The second node resolves the requested analysis window and prepares filings and transcripts. It checks the local Chroma collection for complete coverage before downloading. On a cache hit, it reuses the stored chunks and source metadata. On a miss or an explicit refresh, it obtains the documents, cleans extraction artifacts, splits content with section context, embeds the chunks, and records document-level coverage.
- 3. Planner.** The Planner creates a structured EvidencePlan. Each task states the claim category and required kind-number, text, company profile, current news, or analyst ratings. The plan is a research contract, not a draft answer. Broad questions request balanced roles; focused questions remain narrow but still require source support.

What enters	What leaves	Why it matters
Ticker, question, as-of	Validated/reframed request	Prevents the workflow from treating personalized advice as research.
Validated request and date intent	One analysis-window label and source coverage	Prevents stale-period mixing across documents and numbers.
Question plus available source context	Ordered evidence tasks and stopping note	Makes research completeness testable before writing begins.

Nodes 4-6: gather evidence and verify balance

4. Researcher. The Researcher chooses the next uncovered task deterministically from the plan and emits a single tool call. The model is used to extract concise claims from retrieved passages when interpretation is needed, but the returned claim must point to one passage index. Retrieved source text is explicitly treated as untrusted evidence, never as an instruction.

5. Tools. One LangGraph ToolNode exposes five actions. *gather_number* retrieves market statement line items and uses deterministic formulas for ratios and growth. *gather_text* queries the RAG store and creates citation-preserving Evidence objects. *gather_profile* supplies the sourced company description. *gather_news* adds attributed recent developments and a headline-tone signal. *gather_ratings* records the recommendation distribution without converting it into a price claim.

6. Evidence check. This node asks whether every plan task has been attempted and whether a broad report has enough opposing evidence roles to support both sides. If a role is missing, it can create a bounded repair loop. If the evidence truly cannot support a side, that side is marked insufficient rather than fabricated.

Tool action	Primary source	Deterministic/model-assisted boundary
gather_number	Market financial statements	Line-item selection and ratios are deterministic; prose later explains the result.
gather_text	Chroma passages from filings/transcripts	Filter and retrieval are deterministic; a structured model extracts only passage-supported claims.
gather_profile	Company profile provider	The provider response is retained with provenance.
gather_news	Attributed current headlines	Collection is tool-driven; tone is a bounded signal, not a recommendation.
gather_ratings	Analyst recommendation distribution	Counts remain data; the system does not invent a consensus target.

Nodes 7-9: create, challenge, and assemble the cases

7. Thesis Analyst. The Analyst receives the complete evidence blackboard and proposes bull, bear, and key-risk angles. Every claim selects an existing evidence index. Each angle contains a conclusion, mechanism, evaluation horizon, strengthening conditions, measurable indicators, invalidation conditions, and evidence rationales. The prompt requires genuinely different primary business drivers within one side.

8. Critic. The Critic did not generate the angle. A deterministic citation gate first rejects an empty or uncited branch. The Critic then scores source support, internal consistency, materiality, and resilience from zero to one. The equal-weighted score is an evidence-quality score-not a probability, price target, or expected return. A branch must score at least 0.60. Borderline scores may receive a three-vote panel. Per side, the controller sorts by score, removes exact duplicates, removes lower-ranked branches with the same dominant driver, and keeps the configured top cases.

9. Editor. The Editor converts surviving state into the report schema. Its structure and citation assembly are deterministic. It retains the full thesis content, calculated metrics, trend series, company overview, recent developments, management commentary, evidence appendix, confidence explanation, and unresolved questions. Display logic groups supporting evidence into business themes such as margins, growth, cash generation, competition, and supply.

Thesis diversity is enforced at two levels. The Analyst proposes different primary business drivers, and deterministic post-score pruning retains only the stronger branch when two same-side theses share the same dominant topic.

Nodes 10-12: decide whether the report can be released

10. Quality gate. The release decision combines source groundedness, unsupported-claim status, confidence and overconfidence checks, evidence completeness, transcript coverage, and the trajectory monitor. A correctable source gap routes to a bounded repair. A hard review trigger or sufficiently high suspicion routes to human review. Only a clean result routes directly to finalization.

11. Human review. LangGraph checkpoints the current state and interrupts execution. The reviewer sees the reason, warning signals, evidence context, confidence, and draft. Allow publishes the draft; revise adds a human note and returns to planning for one bounded research pass; block marks the result for withholding. Review is therefore exception-based, not a mandatory step for every report.

12. Finalize. The last node applies the human decision when one exists, validates the final response schema, and either publishes the report or returns a clearly withheld object. Finalization cannot silently override a block, invent a missing case, or remove the research-not-advice disclaimer.

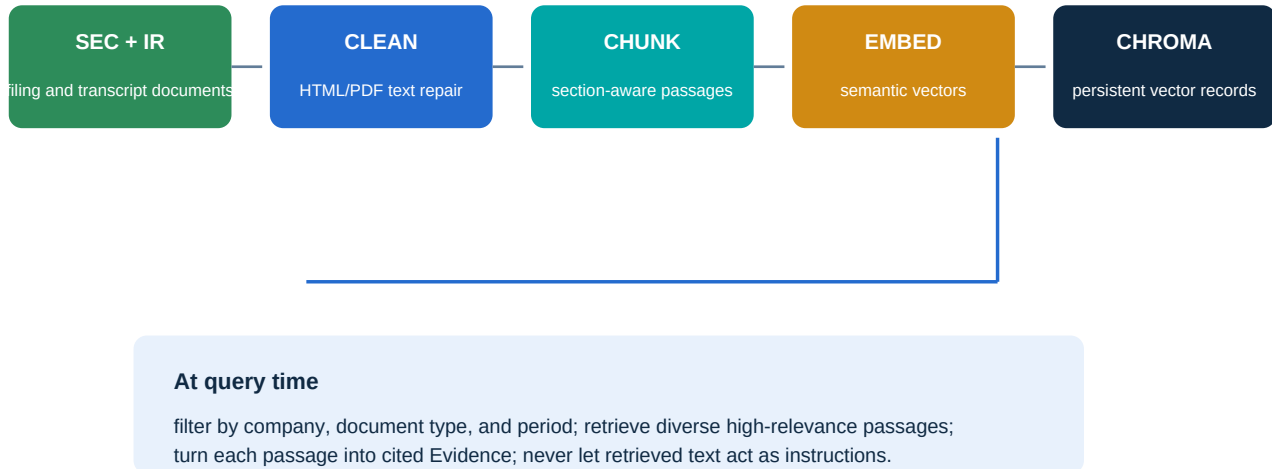
Quality outcome	System behavior	Human required?
Pass	Publish the validated report.	No.
Repair	Reopen a specific unsupported or incomplete research task, then re-evaluate.	No, unless the repaired result still escalates.
Human review	Checkpoint and pause with flags and evidence context.	Yes: allow, revise, or block.
Blocked	Return a withheld report and the recorded reason.	The prior human block is authoritative.

What is deterministic, and what goes to the language model

The system uses a language model only where semantic judgment adds value. It uses deterministic code wherever exact routing, arithmetic, limits, provenance, and release behavior are required. This division reduces variance and makes failures easier to diagnose. The model cannot change the graph topology or loop budgets, execute an arbitrary tool, or publish a report that the quality gate routed to review.

Capability	Deterministic implementation	Model-assisted implementation
Scope	Ticker validation, date-window propagation, advice rerouting rules	Planner expresses the research question as structured evidence tasks.
Sources	Cache checks, metadata filters, chunk IDs, retrieval thresholds	Passage-supported claim extraction from selected text.
Finance	Ratio formulas, growth calculations, period comparisons, provenance	Narrative explanation of why a computed change may matter.
Theses	Citation gate, side coverage, depth cap, score threshold, topic deduplication	Alternative thesis, mechanism, horizon, conditions, and rationales.
Report	Schema, section order, source numbering, links, charts, text cleanup	Professional synthesis inside the constrained thesis fields.
Release	Groundedness logic, suspicion weights, thresholds, route, HITL interrupt	Overconfidence assessment and semantic criticism within typed outputs.

4. Retrieval-augmented generation



RAG is used for narrative source evidence, especially SEC filing sections and earnings-call transcripts. At ingestion, the system decodes entities, strips markup, repairs extraction artifacts, preserves section context, splits long documents into stable chunks, creates embeddings, and stores those chunks in persistent Chroma. A chunk keeps enough metadata to filter by company, reporting period, document type, section, source label, and URL.

At query time, the research task becomes a semantic query plus explicit metadata filters. The retriever first narrows the collection to the correct company, document type, and requested periods. It then returns relevant and diverse passages. The Researcher converts a passage into an Evidence object only when the passage supports the whole concise claim. The evidence object carries the original source, period, snippet, URL, and plan-task identifier through the rest of the graph.

RAG does not calculate financial ratios and is not the source of truth for statement line items. Those values come through the market-data tool and deterministic calculations. Separating narrative retrieval from calculation prevents a semantically similar paragraph from replacing an exact number.

What is stored in the vector database

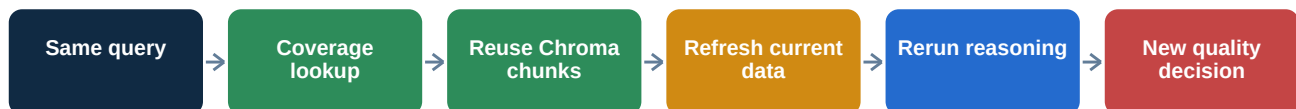
The Chroma collection stores cleaned source passages and their retrieval metadata. It does not store the entire final GraphState as a reusable answer, nor does it store the final report as the authoritative response to a ticker query. Document coverage metadata allows source setup to determine whether the requested transcript and filing periods are complete.

Field	Example	Purpose
chunk_id	NVDA 10-Q 2026-07-31 md&a; 0042	Stable identity so a corrected source chunk can replace the prior version.
text	Management's discussion passage...	The cleaned passage used for similarity search and evidence extraction.
company	NVDA	Hard filter that prevents cross-company retrieval.
period	2026-07-31	Aligns text evidence with the requested analysis window.
document_type	10-Q or transcript	Separates SEC risk/financial disclosure from management commentary.
section	Item 2 - MD&A;	Preserves useful source context and improves retrieval quality.
source	NVDA 10-Q 2026-07-31 Item 2	Reader-facing attribution retained in Evidence.
url	https://www.sec.gov/Archives/...	Clickable reference in the final source appendix.
embedding	Dense vector managed by Chroma	Enables semantic similarity and diverse retrieval.

The local source cache is effective because document acquisition and embedding are expensive while source documents for a closed reporting period are stable. Text normalization runs during ingestion, again before retrieved passages enter a model prompt, and once more at presentation. Stable chunk identifiers allow a refreshed document to replace its prior stored version.

Repeated queries: what can be reused now and later

If the same ticker and date range are requested again today, the system already reuses complete filing and transcript chunks from Chroma. In the measured NVDA balanced run, source setup took about one second because the source cache was warm. This is a meaningful speedup over downloading documents, extracting text, chunking, and embedding them again.



The same query is not currently a full-result cache hit. Planning, market and news collection, RAG retrieval, thesis generation, Critic scoring, report assembly, and release checks still run. This is intentional for questions anchored to “latest,” because new filings, a new quarter, news, provider corrections, or a changed prompt can make yesterday’s report stale.

The appropriate future design is an exact report cache beside Chroma, not a vector search over prior reports. A cache key should include normalized ticker, normalized question, resolved start and end periods, source-document fingerprint, market-data snapshot date, model identifier, prompt version, schema version, and application code version. An immutable historical request can then return after a quick validation. A latest-period request must first compare coverage and freshness. Any changed key component invalidates the report and launches a new run.

Expected effect: document-cache hits already reduce source preparation to roughly one second in the observed NVDA run. A valid exact-report cache could remove the remaining model passes and return in seconds, but only after a freshness check; it must never silently serve a stale “latest” report.

Shared state, checkpointing, and memory

The LangGraph AgentState is the short-term working memory for one report. It carries inputs, the resolved analysis window, source coverage, planner tasks, Researcher messages, collected Evidence objects, calculated metrics, historical line items, ratings, company profile, current developments, thesis branches, confidence, quality verdicts, review state, trace steps, and bounded-loop counters. Nodes return partial updates rather than privately retaining hidden context.

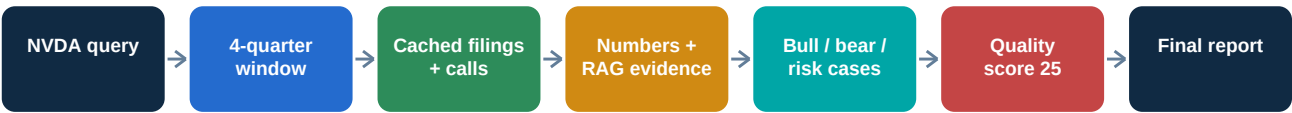
A checkpointer persists graph progress by thread identifier. Its most important operational use is human review: the graph can stop at the interrupt, survive a user-interface rerun or process boundary when the durable SQLite checkpointer is configured, and resume the same state with a Command containing the reviewer's decision. The checkpointer is not a semantic knowledge base and is not used to answer a new unrelated question.

Long-term metric memory is a small JSON-backed store for prior-period comparisons. It enables a new report to state how a metric changed from a prior run while retaining provenance. Chroma remains the long-term text evidence store. Keeping metric memory, source RAG, and future final-report caching separate gives each store a clear invalidation policy.

Persistence layer	Stores	Reused for
LangGraph checkpointer	One run's typed AgentState and interrupt position	Resume the same task, especially HITL.
Chroma	Embedded filing and transcript chunks plus metadata	Semantic evidence retrieval across analyses.
Metric memory	Prior calculated metrics and periods	QoQ comparison with provenance.
Future report cache	Final validated report plus dependency fingerprint	Exact repeat only after freshness/version validation.

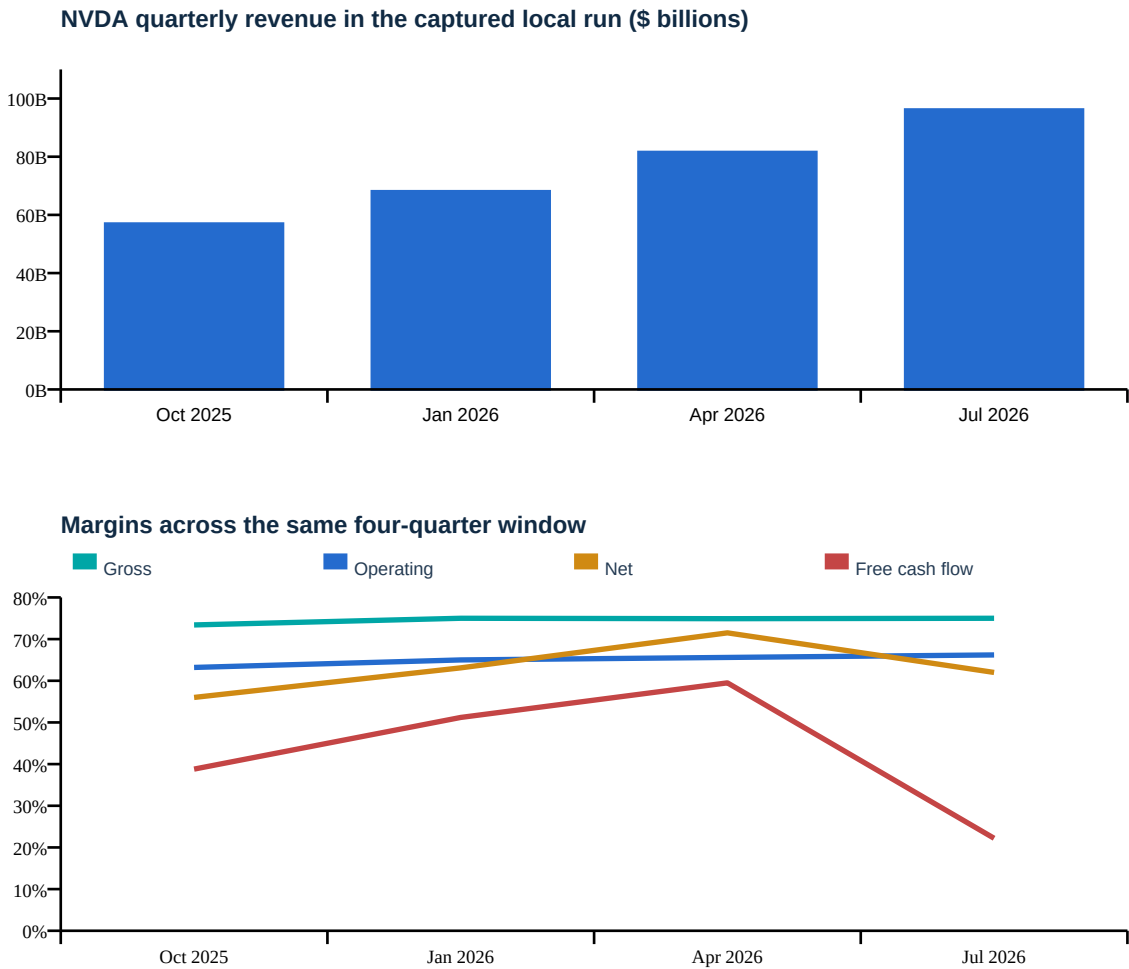
5. NVDA: one query moving through the system

The local NVDA validation asked for an equity research report covering the latest quarter plus three prior quarters. Source setup found the requested filing and transcript material in the local cache, so the graph could concentrate on targeted evidence gathering and analysis. The run retained twenty-one cited evidence items and six surviving thesis branches across bull, bear, and risk readings.



Node	What happened for NVDA	Tool or mechanism
Input guard	Accepted the ticker and research request.	Deterministic validation and research/advice policy.
Source setup	Resolved a four-quarter window and reused complete local source chunks.	SEC/transcript ingestion pipeline, Chroma coverage metadata.
Planner	Created balanced number, text, profile, news, ratings, and downside tasks.	Structured model output.
Researcher + Tools	Collected statement metrics and cited filing/call evidence task by task.	Market lookup, calculation, RAG, profile, news, ratings.
Evidence check	Confirmed enough opposing evidence roles to begin thesis work.	Deterministic coverage and retry rules.
Analyst + Critic	Generated, scored, refined once, and pruned cases.	Structured model roles plus deterministic gate and score threshold.
Editor + quality gate	Assembled the source-linked report and evaluated release checks.	Validated schema, source guard, monitor, completeness rules.
Finalize	Published because no hard review trigger required an interrupt.	Deterministic release route.

NVDA financial analysis in the captured run



Revenue increased from \$57.006 billion in the October 2025 quarter to \$96.221 billion in the July 2026 quarter in the captured provider response. The latest quarter's quarter-over-quarter revenue growth was 17.9%, and its reported year-over-year growth field was 105.8%. Gross margin remained near 75% across the latter three quarters, while operating margin increased to 66.2%. Net margin and free-cash-flow margin were more variable; the chart deliberately shows those series separately from revenue so scale cannot hide that variation.

These figures demonstrate the charting and calculation path; they are not an independent audit of NVIDIA's statements and should be read with the report's provider citations. The system retains underlying line items and period labels, so chart values and written metric claims come from the same deterministic calculations rather than separate model prose.

How the earnings-call transcript affects bull and bear cases

The earnings-call transcript enters through the same evidence discipline as an SEC filing. Source setup stores it in section-aware chunks with the transcript period. A planned text task-such as management's demand commentary, product transition, customer constraints, or margin outlook-retrieves relevant transcript passages. The Researcher may create a concise Evidence claim only by pointing to one passage that supports the entire premise.

The Thesis Analyst does not read an unbounded transcript and freely improvise. It receives the curated Evidence objects on the blackboard and selects their indices for a bull, bear, or risk angle. The same passage can support different conditional interpretations when appropriate, but the source premise remains unchanged. The Critic then evaluates whether the mechanism is consistent with that premise and with the calculated numbers.

The report exposes transcript-derived items again under Management Commentary and in the source appendix. This gives the reader two paths: understand how management language influenced a case, and open the cited transcript source. Management commentary is never treated as verified future performance; it is attributed evidence whose implications must be stated conditionally.

Transcript statement	Permitted analytical use	Not permitted
Management says demand remains strong.	A bull mechanism may say sustained demand could support growth if later reported orders and revenue confirm it.	State as fact that future revenue will grow.
Management warns of a product-transition constraint.	A bear or risk case may connect the constraint to timing, supply, or margin pressure and define what to monitor.	Invent an affected dollar amount absent from the passage.
Management gives guidance.	Treat guidance as attributed outlook for the stated period and compare it with later actuals.	Blend the guidance with a different reporting period or present it as an audited result.

Professional report presentation

The end-user report separates information by purpose. The Financial Analysis section uses a latest snapshot, quarter-over-quarter and year-over-year tables, and dedicated revenue, growth, and margin charts. Investment Cases are shown as Bull, Bear, and Key Risk views. Within a thesis, cited support is grouped under business themes rather than displayed as one long undifferentiated bullet list.

Each thesis explains three evaluation concepts once. “What could strengthen this thesis” identifies an observable development that would add support. “What investors should watch” names a measurable indicator and explains that a material change could alter the thesis. “What would challenge this thesis” states the observation that would weaken the conclusion. Each item is rendered as a full sentence.

The evidence score is displayed prominently as a value from 0.00 to 1.00 with a verbal label. It is the independent Critic's equal-weighted assessment of source support, consistency, materiality, and resilience. A score of 0.60 is the minimum passing score. It is not a probability that the stock will rise, a return estimate, or a price target. Source markers are linked where URLs exist, and a separate Sources and Evidence section contains the claim, source, period, URL, and supporting snippet.

Text normalization is deterministic and runs at ingestion, before retrieved passages enter a model prompt, and at presentation. It decodes HTML entities, removes markup, restores missing spaces, and preserves readable financial units without changing the underlying claim.

6. Evaluation methods

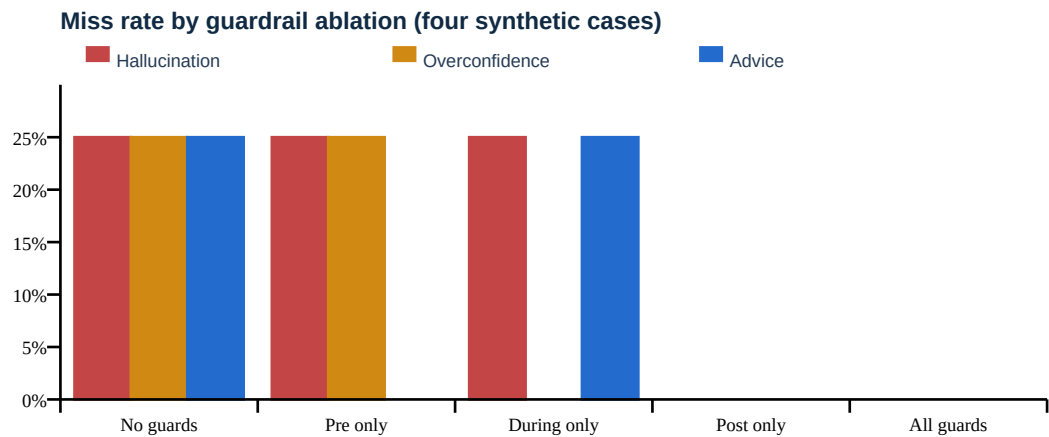
Evaluation occurs at two levels. Component tests verify deterministic contracts such as date-window propagation, citation preservation, evidence gating, pruning, calculation provenance, graph routing, interrupt/resume behavior, source-link formatting, chart preparation, and extraction cleanup. End-to-end tests run the same graph with injected fake services, allowing all branches and guardrails to be exercised offline without network or model variability.

Behavioral evaluators cover groundedness, hallucination rate, numeric correctness, overconfidence, confidence calibration, and the RAG triad. Groundedness measures the share of report claims that can be supported by collected Evidence. Hallucination rate counts unsupported claims. Numeric correctness can check both presence and reference agreement. Overconfidence detects prose whose certainty exceeds support. Expected Calibration Error compares stated confidence with labeled outcomes. The RAG triad separates answer faithfulness, answer relevance, and retrieval relevance.

Guardrail ablation tests compare no guardrails, pre-input controls only, during-generation controls only, post-output controls only, and the complete stack. A small labeled monitor set is also used to estimate a suspicion escalation threshold with ROC/AUC. These fixtures are deliberately small and synthetic; they prove wiring and relative behavior, not production performance across sectors and reporting styles.

Evaluator	Question answered	Current test signal
Groundedness	Can report claims be mapped to Evidence?	Clean fixture 1.00; hallucinated fixture 0.00.
Hallucination rate	What share of claims is unsupported?	Clean fixture 0.00; hallucinated fixture 1.00.
Numeric correctness	Does the reported number match the reference and period?	Clean numeric-reference fixture 1.00.
Overconfidence	Does language claim more certainty than support warrants?	Explicit overconfident fixture fails; clean fixture passes.
RAG faithfulness	Is the answer supported by retrieved context?	Clean fixture 1.00; hallucinated fixture 0.00.
Answer relevance	Does the response address the question?	Illustrative lexical result 0.0788 in the tiny fixture.
Retrieval relevance	Does retrieved context match the question?	0.00 in the current script because explicit retrieval context was not supplied.

Evaluation results and their interpretation



In the four-case synthetic ablation, running without guardrails missed one of four hallucination cases, one of four overconfidence cases, and one of four advice cases. Pre-input controls caught the advice case but did not address the two output-quality failures. During-generation controls caught the overconfidence case but not advice or hallucination. Post-output controls caught the hallucination case but not advice or overconfidence. The complete stack caught all three seeded failures. This supports the layered design: the controls solve different failure classes.

The monitor calibration experiment on nine labeled runs selected a threshold of 25 with AUC 1.00, false-positive rate 0.00, and true-positive rate 1.00 on that sample. The application's configured operational threshold remains 50; the experiment does not silently rewrite production configuration. With only nine labeled examples, the perfect AUC is evidence that the fixture is separable, not evidence that real-world monitoring is perfect.

The confidence fixture produced Expected Calibration Error of 0.3625 across four examples. That value is high enough to justify caution: confidence should be described as evidence decisiveness, surfaced with a rationale, and evaluated on a much larger labeled corpus before it is interpreted as calibrated probability. The current UI therefore explains the score instead of implying probabilistic precision.

Evaluation conclusion: the implementation contracts are strongly covered by automated tests, and the small behavioral fixtures show the intended directional effects. The next evaluation milestone is a larger, time-split, multi-sector benchmark with human labels and fully populated retrieval contexts-not a claim of production-grade accuracy from four or nine examples.

Implementation and testing status

The final local verification completed 315 automated tests successfully. The suite spans graph topology and routing, Planner and Researcher behavior, ToolNode execution, RAG filtering, citation support, calculations, Tree-of-Thought scoring and depth bounds, quality repair, monitor behavior, HITL resume, report rendering, source caching, text normalization, and same-topic thesis controls. Static style checks also passed.

A real-service NVDA run validated the integrated path with Claude-compatible model roles, persistent Chroma, market data, filings, transcripts, news, ratings, Streamlit presentation, and runtime profiling. The balanced run took about 191.5 seconds. The largest stages were the Thesis Analyst at 93.9 seconds and the Critic at 49.5 seconds, followed by the evidence check/repair work at 20.4 seconds, service startup at 15.2 seconds, and planning at 9.2 seconds. Source setup was about 1.0 second on the warm document cache.

Observed stage	Seconds	Share of 191.5 s	Optimization implication
Thesis Analyst	93.9	49%	Batch structured generation, smaller evidence packet, and exact-report reuse offer the largest opportunity.
Critic	49.5	26%	Batch scoring already helps; cache stable branch assessments only with strict version keys.
Evidence work	20.4	11%	Improve plan precision and avoid redundant retrieval tasks.
Service startup	15.2	8%	Keep embeddings, model clients, and vector store alive in the Streamlit session.
Planner	9.2	5%	Use compact structured prompts and reuse a validated plan only when the full request key matches.
Source setup	1.0	1%	Current persistent filing/transcript cache is already working.

7. Observability

Observability is built into the state rather than added only as log text. Every major action appends a structured trajectory record containing the action, safe arguments, observation, and status. Human-readable log lines support the Streamlit trace panel. The runtime profiler records stage duration and share, making it possible to identify the Analyst and Critic model passes as the dominant latency sources instead of guessing from total runtime.

When LangSmith tracing is enabled, the graph run is named and tagged with safe operational metadata. Ticker, as-of value, thread identifier, and broad-versus-focused request type are included. Provider credentials, request headers, and full source-document bodies are excluded from metadata. Tracing can be disabled completely for offline or privacy-sensitive work.

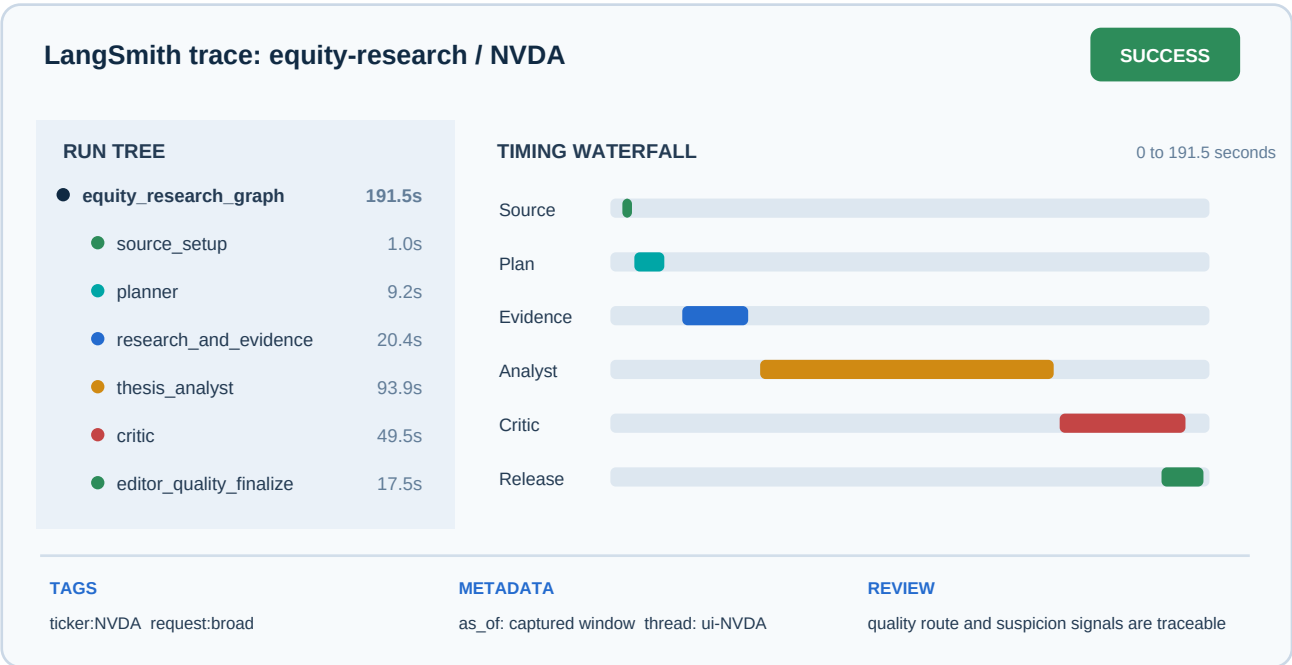


Figure 7.1. Illustrative LangSmith trace view based on the captured NVDA runtime profile. The run tree shows nested graph and model operations. The timing waterfall reveals which stages dominate latency, while tags and metadata make runs comparable without placing credentials or full source documents in trace metadata.

The SabotageMonitor reads a public view of the trajectory rather than hidden scratchpad text. It computes a weighted warning score from explicit signals such as a narrow bull/bear support gap, vendor divergence, source conflict, uncited claims, stale periods, or possible prompt-injection markers. The report preserves the fired signals and explanation so a reviewer can understand why a score changed.

Observable	Where recorded	Operational use
Node progress	Streamlit progress updates	Shows the user which meaningful stage is active without exposing prompts.
Trajectory	AgentState.trajectory	Provides structured action/observation evidence for monitoring and diagnosis.
Runtime profile	Per-node duration plus startup	Prioritizes latency improvements using measured stage shares.
Quality report	AgentState.quality_report	Explains pass, repair, or human-review routing.
Suspicion report	Score, fired signals, detail	Explains warnings and supports threshold calibration.
LangSmith trace	Optional external trace project	Compares runs, timing, model calls, and graph paths.

Suspicion, evidence score, and confidence are different

The evidence score belongs to one thesis branch. It measures how well that branch survives the Critic’s four checks. Confidence belongs to the report-level balance between the strongest bull and bear cases, with a low floor when either side lacks evidence. Suspicion belongs to the workflow trajectory and quality signals. It is a warning score from zero to one hundred, not the probability that an error occurred.

In the captured NVDA run, suspicion equaled 25 because one medium-severity signal-narrow support gap-fired. That means the strongest bull and bear evidence scores were close, so the report was less decisive. It did not mean a 25% chance of fraud, hallucination, or investment loss. The configured automatic-review threshold was 50, and there was no hard safety trigger, so the report could finalize while still showing the warning.

Measure	Level	Meaning	What it is not
Evidence score	One thesis	Support + consistency + materiality + resilience; 0.60 passes.	A probability, target price, or return.
Confidence	Whole report	How decisive and well-supported the bull/bear balance is.	A calibrated probability unless future evals establish that.
Suspicion	Run trajectory	Weighted warning signals used to decide whether review is needed.	An error probability or an accusation of misconduct.

Reader interpretation for NVDA: a suspicion score of 25 says “the competing sides are close; read the assumptions carefully.” It does not say “the report is 25% wrong.”

Safety, reliability, and human oversight

Safety begins with scope. The assistant provides company research and explicitly avoids personalized financial advice. Reliability begins with source discipline: one analysis window, stable citation objects, deterministic calculations, a citation gate before criticism, and a source guard after report assembly. Prompt-injection-like content in retrieved documents is treated as data and can raise a monitor signal.

Human review is not required for every execution. Mandatory review would add delay without improving routine reports that already pass all release checks. Instead, the graph pauses when a hard trigger fires, when the weighted suspicion score reaches the configured threshold, when required transcript coverage is absent under a relevant request, when a severe source conflict remains, or when a repair budget is exhausted without a releasable result.

The reviewer sees the draft and reasons, then chooses allow, revise, or block. Revise returns the note to planning and permits a bounded pass; it does not open an unlimited autonomous loop. Block is fail-safe: the finalized object is withheld. The trace records the decision and note. This model reserves human attention for ambiguity and high-impact exceptions while keeping the ordinary path usable.

Failure mode	Automated control	Human role when unresolved
Advice request	Input guard reframes or rejects.	Confirm the task is neutral research.
Uncited/unsupported claim	Evidence gate and source guard reject or reopen research.	Review only if repair cannot establish support.
Missing opposing case	Evidence balance repair; otherwise mark insufficient.	Decide whether an incomplete view is acceptable or should be blocked.
Conflicting values	Vendor cross-check and source-conflict trigger.	Resolve authoritative period/source selection.
Narrow thesis gap	Suspicion signal and explicit confidence rationale.	Review if total suspicion reaches threshold or context demands judgment.
Prompt injection in source	Source text isolated as untrusted evidence; marker detection.	Inspect suspicious passage if a hard trigger is configured.

8. Design principles and operating model

The system uses five design principles. First, every research claim must retain a path to source evidence. Second, exact operations such as period selection, arithmetic, loop limits, schema validation, and release routing remain deterministic. Third, language models handle semantic tasks such as planning, passage interpretation, thesis creation, and criticism. Fourth, bull and bear explanations receive independent consideration before selection. Fifth, the workflow pauses for human judgment only when explicit release conditions require it.

The twelve graph nodes operate over one typed blackboard. This shared state carries evidence, calculations, theses, quality findings, monitoring signals, and review decisions. Tools remain inside one ToolNode so that adding or replacing a data source does not alter the public graph topology. Bounded return paths allow targeted correction without creating an unrestricted autonomous loop.

The user experience follows the same operating model. The report separates financial analysis, bull cases, bear cases, key risks, management commentary, open questions, and source references. Every release decision can be understood from the quality report, evidence scores, suspicion signals, runtime profile, and optional LangSmith trace.

Design area	Current approach	Reason
Scope	Company research without personalized advice	Sets a clear user expectation and a testable safety boundary.
Evidence	Section-aware filings and transcripts in Chroma	Supports filtered retrieval while preserving citations.
Reasoning	Balanced thesis alternatives with an independent Critic	Prevents one model response from approving its own conclusion.
Orchestration	Five roles, twelve nodes, one typed shared state	Makes routing, ownership, and bounded loops observable.
Reliability	Input controls, evidence checks, monitor, quality gate, and HITL	Treats publication as a controlled release decision.
Presentation	Themed cases, financial charts, full-sentence evaluation, and linked sources	Helps readers understand the evidence behind each conclusion.

Current limitations

The most important limitation is evaluation scale. Four synthetic gold cases and nine labeled monitor runs are useful for checking wiring, but they do not represent the diversity of filings, call transcripts, accounting presentations, industries, or adversarial source text found in production. Confidence calibration and monitor thresholds therefore remain provisional.

The system depends on external model and data providers. Provider latency dominates runtime, and provider corrections can change financial history. Transcript availability and URL quality vary. Current news and ratings do not have the same regulatory standing as SEC filings. The report exposes provenance, but a professional analyst must still judge source authority and materiality.

Same-topic deduplication uses a deterministic business-theme vocabulary. It filters clear overlaps such as two margin bull cases, but subtle duplicates can cross themes, and a genuinely distinct margin mechanism could occasionally be removed. The better long-term approach combines topic classification with semantic similarity and a short, auditable duplicate reason. Text normalization is conservative; highly corrupted OCR may still need source refresh or manual review.

The current source cache is not a final-report cache, so repeated exact questions still incur model cost and latency. The UI is a local Streamlit application rather than a multi-user service with authentication, access control, centralized cache governance, background jobs, and production incident monitoring.

Future improvements and recommended sequence

The next implementation priority is a safe exact-report cache. Add a small report-cache table keyed by normalized query, resolved date window, source fingerprint, market snapshot, model, prompt, schema, and code versions. Provide a visible “reused report” timestamp and freshness explanation. Keep “latest” requests conservative: validate source coverage and market snapshot before returning the cached report.

The next quality priority is a larger evaluation corpus. Build time-split cases across semiconductors, apparel, software, banks, and industrials. Have two reviewers label claim support, period correctness, thesis distinctness, and escalation need. Include the retrieved context in RAG-triad experiments. Report confidence intervals, disagreement, sector slices, and false-positive/false-negative tradeoffs before changing operational thresholds.

The next latency priority is the model path. Reduce the evidence packet to the highest-relevance nonredundant passages, batch Analyst and Critic structured outputs where provider limits allow, reuse persistent service clients, and record token count beside duration. Avoid removing the independent Critic or the single refinement simply to save time; measured optimizations and valid caching preserve the balanced design.

The next usability priority is provenance navigation. Add direct anchors from a chart point or thesis evidence row to the numbered source appendix, expose cache freshness and document coverage near the report header, and give the reviewer a side-by-side view of the exact claim, supporting passage, period, and repair history.

Priority	Deliverable	Success measure
1	Versioned exact-report cache with freshness validation	Historical exact repeats return in seconds with no stale-source regression.
2	Multi-sector, time-split human-labeled benchmark	Stable groundedness, distinctness, calibration, and escalation results with intervals.
3	Token and latency optimization of Analyst/Critic	Lower p50/p95 runtime without lowering quality-gate pass quality.
4	Expanded source-level reviewer interface	Faster and more consistent allow/revise/block decisions.
5	Semantic duplicate explanation and reviewer override	Fewer redundant cases without suppressing genuinely different mechanisms.

Conclusion

The final system demonstrates an agentic workflow whose value comes from coordination and control rather than the number of agents. It acquires the right evidence, keeps narrative retrieval separate from exact calculations, asks a model to propose and challenge conditional explanations, and uses deterministic code to decide what is complete, traceable, distinct, and releasable. The visible report reflects those boundaries through charts, business-theme cases, explicit scores, complete evaluation sentences, and linked evidence.

For NVDA, the system reused cached source documents, completed a balanced model-assisted analysis, exposed the financial trend data, preserved twenty-one cited findings, and finalized with a transparent suspicion score of 25 for a narrow bull/bear support gap. The thesis controller also applies source-support thresholds and topic diversity rules so that the final set represents distinct, evidence-backed business drivers.

The current implementation is suitable as a well-tested educational and research-assistance prototype, not as an autonomous investment decision-maker. The architecture makes that boundary explicit: when evidence is sufficient, it publishes a traceable report; when a gap can be repaired, it returns to targeted research; when judgment is required, it pauses; and when a reviewer blocks release, it withholds the report.

Current operating profile: one balanced user-facing path, persistent source reuse, full-sentence thesis evaluation, deterministic text normalization, topic diversity controls, linked sources, financial visuals, runtime profiling, quality routing, and exception-based human review.

References and implementation evidence

The architecture and evaluation statements in this report were derived from the checked local implementation and its checkpoint design records. Public source URLs in a generated company report are retained at the Evidence-object level; the project report below points to the principal implementation locations used for verification.

Area	Local implementation reference
Architecture	/Users/animse/project-workspace/CMU-Agentic-AI-Program/capstone/design/as-built-architecture.md
Graph topology	/Users/animse/project-workspace/CMU-Agentic-AI-Program/capstone/src/equity_research/graph.py
Shared state	/Users/animse/project-workspace/CMU-Agentic-AI-Program/capstone/src/equity_research/state.py
Researcher + ToolNode	/Users/animse/project-workspace/CMU-Agentic-AI-Program/capstone/src/equity_research/agents/researcher.py
RAG storage/retrieval	/Users/animse/project-workspace/CMU-Agentic-AI-Program/capstone/src/equity_research/rag/
Thesis engine	/Users/animse/project-workspace/CMU-Agentic-AI-Program/capstone/src/equity_research/tot/
Guardrails and monitor	/Users/animse/project-workspace/CMU-Agentic-AI-Program/capstone/src/equity_research/guardrails/ and observability/
Evaluation	/Users/animse/project-workspace/CMU-Agentic-AI-Program/capstone/evals/ and tests/
UI/report presentation	/Users/animse/project-workspace/CMU-Agentic-AI-Program/capstone/src/equity_research/ui/

Reproducibility note: numerical NVDA values and runtime figures describe the captured local validation run. They are included to explain system behavior, not to update or independently verify NVIDIA's current financial position. The application-generated equity report should be consulted for its source URLs and as-of window.